

KUSH PATEL

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Education

McMaster University

B.A.Sc - Computer Science (Minor in Statistics)

Expected Grad. May 2028

Hamilton, ON

Experience

Royal Bank of Canada (RBC)

Jan 2026 – Present

Software Developer Intern – Enterprise Architecture

Toronto, ON

- Eliminated stale documentation and **saved 26k+ hours** of manual documentation YTD across **1,316 unique repos** by leveraging LLMs to analyze codebases and generate system architecture diagrams and full technical wikis
- Accelerated multi-repo analysis speed by **100%** and **doubled capacity** from 10 to 20 concurrent repos per analysis, by engineering a parallel **multi-agent orchestration** system
- Decoupled a legacy UI-locked monolith into independent **microservices**, exposing the core platform through a single API to enable **MCP** and **CI/CD** adoption allowing fresh documentation to spin up on every production release
- Drove a company-wide Claude Code rollout achieving **100% business-line adoption**, by single-handedly building a **full-stack governance portal** featuring 3-tier RBAC, audit trails, and spend analytics for **VP-level ROI** tracking.

Moriroku Technology North America

June 2025 – Aug 2025

Software Engineering Intern

Listowel, ON

- Automated Production Scheduling by building a **live order data pipeline** from MS D365 to MS SQL to a **CP-SAT model (Google OR-Tools)** delivering a Gantt Chart in Vue, expanding scheduling capacity by **3-4x** per cycle
- Delivered a real-time Vue OEE dashboard, boosting downtime response speed by **~30%** across 20+ production lines
- Designed** a centralized template management platform (PostgreSQL + Node.js/Express + FastAPI) with auth and audit logging, cutting update time from **hours to minutes** and reducing errors by **80%**

PBJ Cleaning Depot Inc.

May 2024 – Aug 2024

Software Engineering Intern

Listowel, ON

- Developed a multi-module ERP system, projected to save **~\$25K/year** annually by reducing external licenses by 80%
- Reduced manual data entry by 60%** and improved inventory **accuracy to 99.5%** by automating stock monitoring and purchase order workflows through real-time tracking
- Built **RESTful APIs** in Go with **JWT-based auth**, bcrypt, and RBAC across **20+ protected endpoints**

Projects

Mesh – GeoSpatial 3D Coordination Platform | *Next.js, React, Three.js, TypeScript*

- Architected a **multi-stage AI pipeline** chaining **Gemini Pro** and **GPT-4** via OpenRouter to auto-segment and annotate **100+ mesh components** per 3D model, eliminating hours of manual CAD labeling
- Built a **real-time WebGL engine** with **Three.js** and **React Three Fiber**, delivering exploded views, component isolation, bloom post-processing, and raycasting selection at **60fps**
- Engineered **BLE hardware integration** with **Arduino M5StickCPlus2**, streaming quaternion data at **500Hz** with **<2ms** latency using a **Madgwick AHRS filter** for physical 3D model control

Mini-IPRO | *Python, PyRadiomics, lifelines* | *In Progress*

- Toy-scale replica of a production oncology-imaging pipeline: **DICOM QC** → radiomic feature extraction → **survival modeling** on public lung-cancer data (TCIA)

ResDex | *Startup Software Lead*

- Leading a team of 5 to build a student-focused **social media platform** for academic research, enabling sharing via QR-enabled portfolios and short-form content
- Collaborated with **12+ PhD professionals**, securing **200+ pre-signups** from top institutions (UofT, McMaster, etc)

Technical Skills

Languages: Python, TypeScript, JavaScript, Go, SQL, HTML/CSS

Libraries: OR-Tools, PyRadiomics, lifelines, Pandas, Scikit-learn, Matplotlib, Three.js, React Three Fiber, Tailwind CSS

Frameworks: Next.js, React, Vue.js, Node.js/Express, FastAPI

Developer Tools: Git, Docker, VS Code, Cursor, Codex, Claude Code, MCP